

Area & Perimeter — Practice Questions

Digital SAT Style | Mixed Difficulty | Total Questions: 25

Q1.

DIFFICULTY: EASY

A rectangle has length 12 cm and width 5 cm. What is its area?

- A) 17 cm²
 - B) 34 cm²
 - C) 60 cm²
 - D) 120 cm²
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Q2.

DIFFICULTY: EASY

A square has a perimeter of 48 cm. What is its area?

- A) 12 cm²
 - B) 96 cm²
 - C) 144 cm²
 - D) 192 cm²
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Q3.

DIFFICULTY: EASY

A triangle has a base of 10 cm and a height of 6 cm. What is its area?

- A) 16 cm²
 - B) 30 cm²
 - C) 60 cm²
 - D) 120 cm²
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Q4.

DIFFICULTY: EASY

A circle has a radius of 7 cm. What is its area? (Use $\pi \approx 22/7$)

- A) 22 cm²
 - B) 44 cm²
 - C) 154 cm²
 - D) 308 cm²
-

Q5.

DIFFICULTY: EASY

A rectangle has length $(x + 3)$ and width 4. Its perimeter is 30. What is x ?

- A) 4

- B) 5
 - C) 7
 - D) 8
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Q6.

DIFFICULTY: EASY

Two squares have side lengths 3 cm and 5 cm. What is the difference in their areas?

- A) 2 cm²
 - B) 4 cm²
 - C) 8 cm²
 - D) 16 cm²
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Q7.

DIFFICULTY: MEDIUM

A trapezoid has parallel sides of 8 cm and 12 cm, and a height of 5 cm. What is its area?

- A) 40 cm²
 - B) 50 cm²
 - C) 60 cm²
 - D) 100 cm²
-

Q8.

DIFFICULTY: MEDIUM

A circle has diameter 20 cm. What is its circumference? (Use $\pi \approx 3.14$)

- A) 31.4 cm
 - B) 62.8 cm
 - C) 125.6 cm
 - D) 314 cm²
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Q9.

DIFFICULTY: MEDIUM

If the radius of a circle is doubled, by what factor does the area increase?

- A) 2
 - B) 4
 - C) 6
 - D) 8
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Q10.

DIFFICULTY: MEDIUM

A rectangular garden is 15 m by 10 m. A path 2 m wide surrounds the garden. What is the area of the path alone?

- A) 56 m^2
 - B) 92 m^2
 - C) 116 m^2
 - D) 124 m^2
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Q11.

DIFFICULTY: MEDIUM

A semicircle has a diameter of 14 cm. What is its area? (Use $\pi = 22/7$)

- A) 38.5 cm^2
 - B) 77 cm^2
 - C) 154 cm^2
 - D) 308 cm^2
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Q12.

DIFFICULTY: MEDIUM

A sector of a circle has a central angle of 90° and a radius of 10 cm. What is the area of the sector?

- A) $25\pi \text{ cm}^2$
 - B) $50\pi \text{ cm}^2$
 - C) $75\pi \text{ cm}^2$
 - D) $100\pi \text{ cm}^2$
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Q13.

DIFFICULTY: MEDIUM

A ring (annulus) is formed between two concentric circles with radii 5 cm and 3 cm. What is the area of the ring?

- A) $8\pi \text{ cm}^2$
 - B) $16\pi \text{ cm}^2$
 - C) $25\pi \text{ cm}^2$
 - D) $34\pi \text{ cm}^2$
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Q14.

DIFFICULTY: HARD

If each side of a square is increased by 20%, by what percent does the area increase?

- A) 20%
 - B) 40%
 - C) 44%
 - D) 120%
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Q15.

DIFFICULTY: HARD

A farmer has 120 meters of fencing to enclose a rectangular area against a barn wall (fencing needed on only 3 sides). What is the maximum area that can be enclosed?

- A) 900 m^2
 - B) 1200 m^2
 - C) 1500 m^2
 - D) 1800 m^2
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Q16.

DIFFICULTY: HARD

A square has an area of 196 cm^2 . A circle is inscribed in the square. What is the area of the circle? (Use $\pi \approx 22/7$)

- A) 77 cm^2
 - B) 154 cm^2
 - C) 196 cm^2
 - D) 616 cm^2
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Q17.

DIFFICULTY: HARD

A right triangle has legs 9 cm and 12 cm. What is its perimeter?

- A) 21 cm
 - B) 30 cm
 - C) 36 cm
 - D) 45 cm
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Q18.

DIFFICULTY: HARD

The area of a triangle with vertices (0, 0), (8, 0), and (4, 6) is:

- A) 16
 - B) 24
 - C) 32
 - D) 48
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Q19.

DIFFICULTY: HARD

A rectangle has a perimeter of 40 cm and a length that is 3 times its width. What is the area?

- A) 48 cm^2
 - B) 75 cm^2
 - C) 100 cm^2
 - D) 120 cm^2
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Q20.

DIFFICULTY: HARD

An equilateral triangle has a side of 8 cm. What is its area?

- A) $12\sqrt{3} \text{ cm}^2$
 - B) $16\sqrt{3} \text{ cm}^2$
 - C) $24\sqrt{3} \text{ cm}^2$
 - D) $32\sqrt{3} \text{ cm}^2$
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Q21.

DIFFICULTY: HARD

A circular garden has radius 10 m. A walking path of width 2 m surrounds it. What is the area of the path?

- A) $40\pi \text{ m}^2$
 - B) $44\pi \text{ m}^2$
 - C) $56\pi \text{ m}^2$
 - D) $100\pi \text{ m}^2$
-

Q22.

DIFFICULTY: HARD

A rectangle has area 48 cm^2 and perimeter 28 cm. What are its dimensions?

- A) 4×12
 - B) 6×8
 - C) 3×16
 - D) 2×24
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Q23.

DIFFICULTY: HARD

A regular hexagon has a side length of 6 cm. What is its area? (Use formula: $A = (3\sqrt{3}/2)s^2$)

- A) $27\sqrt{3} \text{ cm}^2$
 - B) $36\sqrt{3} \text{ cm}^2$
 - C) $54\sqrt{3} \text{ cm}^2$
 - D) $72\sqrt{3} \text{ cm}^2$
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Q24.

DIFFICULTY: HARD

A square and a circle have the same perimeter of 44 cm. Which has the larger area and by how much? (Use $\pi = 22/7$)

- A) Circle, by approximately 33 cm^2
- B) Square, by approximately 33 cm^2
- C) Circle, by approximately 56 cm^2

D) Square, by approximately 56 cm^2

Q25.

DIFFICULTY: HARD

A parallelogram has base 14 cm and height 9 cm . A triangle with the same base and height is cut from it. What is the remaining area?

- A) 63 cm^2
 - B) 72 cm^2
 - C) 81 cm^2
 - D) 126 cm^2
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